

DAWEI BAI

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Updated 6/2026

ACADEMIC POSITIONS

Postdoc , Yale University, New Haven, CT, USA. Sponsors: Brian Scholl and Samuel McDougle	2023–Present
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EDUCATION

PhD , École Normale Supérieure, Institut Jean Nicod, Paris, France. Supervisor: Brent Strickland	2019–2023
M.Sc., Cognitive Science (“Cogmaster”), École Normale Supérieure, École des Hautes Études en Sciences Sociales and Paris Descartes University, Paris, France.	2017–2019
M.A., Linguistics , University of Lille, Lille, France.	2016–2017
B.S., Psychology , University of Lille and University of Poitiers, Lille, France.	2014–2017
B.A., French Literature , University of Poitiers, Poitiers, France.	2013–2016
Engineering study , University of Technology of Compiègne, Compiègne, France.	2011–2013

PUBLICATIONS

- Bai, D.**, Hafri, A., Izard, V., Firestone, C., & Strickland, B. (2025). “Core perception”: Re-imagining precocious reasoning as sophisticated perceiving [target article]. *Behavioral and Brain Sciences*. <https://doi.org/10.1017/S0140525X25102756>
- Bai, D.**, & Strickland, B. (2025). The “Double Ring Illusion”: The physical constraint of solidity shapes visual processing. *Journal of Experimental Psychology: General*, 154(12), 3417–3427. <https://doi.org/10.1037/xge0001839>
- Bai, D.**, & Strickland, B. (2023). The Pulfrich solidity illusion: A surprising demonstration of the visual system’s tolerance of solidity violations. *Psychonomic Bulletin & Review*, 30(5), 1782–1787. <https://doi.org/10.3758/s13423-023-02271-9>

PAPERS UNDER REVIEW

Bai, D., & Scholl, B. J. (under revision). The tiger moving through the tall grass: Spontaneous perception and tracking of change-defined objects from second-order motion.

Bai, D., Weisman, K., Schille-Hudson, E., Ihm, E., Taves, A., Luhrmann, T. M., & Scholl, B. J. (under revision). Chasing the supernatural: The perception of animacy from motion is related to spiritual experiences.

WORKING PAPERS

Bai, D., Ongchoco, J. D. K., Weisman, K., Schille-Hudson, E., Ihm, E., Taves, A., Luhrmann, T. M., & Scholl, B. J. (in preparation). From grids to gods?: Perceiving visual structure via ‘scaffolded attention’ is related to spiritual experiences.

Bai, D., & McDougle, S., (in preparation). Reaching under (visually) inferred wind forces.

Bai, D., Dhar, P., Wong, K., & Scholl, B., (in preparation). Do visual objects lose their individuality due to the perception of collective goals?: Evidence from numerical underestimation in the Wolfpack effect.

Bai, D., & Scholl, B. J., & Strickland, B. (in preparation). Isolating event categories in space and time: Evidence of dynamic “on the fly” representations in perception.

Jones, H., **Bai, D.,** Scholl, B. J., & Awh, E. (in preparation). Electroencephalogram decoding of the attentional selection and tracking of featureless objects.

O'Madagain, C., **Bai, D.,** Benmarrakchi, F., Tetteh, G., Peperkamp, S., Strickland, B. (in preparation). It is easier to learn a new grammar for core concepts.

Sun, Z., **Bai, D.,** Scholl, B. J., & McDougle, S. (in preparation). Visuomotor adaptation via featureless objects.

INVITED TALKS

2026 “Core perception”: Re-imagining precocious reasoning as sophisticated perceiving.
Ghislaine Dehaene-Lambertz lab, NeuroSpin, France.

2026 Chasing the supernatural: The perception of animacy from motion is related to spiritual experiences. Melissa Võ lab, Ludwig Maximilian University of Munich.

2026 “Core perception”: Re-imagining precocious reasoning as sophisticated perceiving.
Collège de France Mind & Language online seminar.

2026 “Core perception”: Re-imagining precocious reasoning as sophisticated perceiving.
Harvard Research Seminar Series, Harvard University.

- 2025 “Core perception”: Re-imagining precocious reasoning as sophisticated perceiving.
Chaz Firestone lab, Johns Hopkins University.
- 2025 The tiger moving through tall grass: Spontaneous perception and tracking of featureless objects.
Martin Rolfs lab, Humboldt-Universität zu Berlin.
- 2023 Intuitive physics in visual perception.
Hans Op de Beeck lab, KU Leuven.
- 2023 From object perception to object cognition: Philosophical and empirical perspectives.
Memory and Mental Files workshop, Grenoble, France.
- 2021 The solidity principle in visual perception.
Language, Thought, and Perception Workshop, Nantes, France.

SELECTED CONFERENCE PRESENTATIONS

- Bai, D.**, Ongchoco, J. D. K., Weisman, K., Schille-Hudson, E., Ihm, E., Taves, A., Luhrmann, T. M., & Scholl, B. J. (2026). From grids to gods?: Perceiving visual structure via ‘scaffolded attention’ is related to spiritual experiences. Poster presented at the *Vision Sciences Society Annual Meeting*, St. Pete Beach, Florida, USA.
- Bai, D.**, & McDougale, S., (2026). Reaching under (visually) inferred wind forces. Poster presented at the *Neural Control of Movement Annual Meeting*, Kobe, Japan.
- Bai, D.**, Hafri, A., Izard, V., Firestone, C., & Strickland, B. (2026). “Core perception”: Re-imagining precocious reasoning as sophisticated perceiving. Talk given at the *Budapest CEU Conference on Cognitive Development*, Budapest, Hungary.
- Bai, D.**, Weisman, K., Schille-Hudson, E., Ihm, E., Taves, A., Luhrmann, T. M., & Scholl, B. (2025). Chasing the supernatural: The perception of animacy from motion is related to spiritual experiences. Poster presented at the *Vision Sciences Society Annual Meeting*, St. Pete Beach, Florida, USA.
- Bai, D.**, & Scholl, B., (2024). Object tracking without objects: Perceiving persistence defined by pure change. Poster presented at the *European Conference on Visual Perception*, Aberdeen, Scotland.
- Bai, D.**, & Scholl, B., (2024). Do visual objects lose their individuality due to the perception of collective goals?: Evidence from numerical underestimation in the Wolfpack effect. Poster presented at the *Vision Sciences Society Annual Meeting*, St. Pete Beach, Florida, USA.
- Bai, D.**, & Strickland, B. (2023). Object solidity disambiguates ambiguous motion. Talk given at the *Vision Sciences Society Annual Meeting*, St. Pete Beach, Florida, USA.
- Bai, D.**, & Strickland, B. (2022). The double ring illusion: Object solidity is used to disambiguate ambiguous motion. Talk and demo night presentation given at the *European Conference on Visual Perception*, Nijmegen, the Netherlands.

- Bai, D., & Strickland, B. (2022).** The double ring illusion: Object solidity is used to disambiguate ambiguous motion. Poster presented and demo night presentation at the *Vision Sciences Society Annual Meeting*, St. Pete Beach, Florida, USA.
- Bai, D., Falck, A., & Strickland, B. (2021).** The adult visual system resists learning continuity violations. Poster presented at the *Virtual Vision Science Society Meeting*. Online.
- Bai, D., O'Madagain, C., Peperkamp, S., & Strickland, B. (2019).** Core Cognition and Learnability: A Possible Explanation for Cross-linguistic Similarities. Talk given at the *Conference of the European Society for Cognitive Psychology*, Tenerife, Spain.

AWARDS

- “Best Illusion of the Year” 2021 (Neural Correlate Society): Third place with the “Double Ring Illusion”.
View the illusion at <https://youtu.be/bVbPmFpspXc>.